

Biswajit Sahoo, Ph.D.

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RESEARCH INTERESTS	Machine Learning, Deep Learning, Natural Language Processing, Agentic Systems, Remaining Useful Life (RUL) Prediction, Exploratory Data Analysis, Signal Processing, Machinery Fault Diagnosis, Internet of Things
EDUCATION	Indian Institute of Technology - Kharagpur, West Bengal, India 2024 Ph.D., Mechanical Engineering Thesis: Data-driven diagnosis and Prognosis of faults in rotating machinery National Institute of Technology - Rourkela, Odisha, India 2015 M. Tech, Mechanical Engineering — GPA: 9.29/10 Specialization: Machine Design and Analysis Odisha University of Technology and Research - Bhubaneswar, Odisha, India 2013 B. Tech., Mechanical Engineering — GPA: 8.09/10
EXPERIENCE	HP Inc. R&D - Bengaluru, India 11/2021 - Present <i>Senior Machine Learning Engineer</i> Through TEKsystems: 11/2021 - 4/2023, Full-time: 4/2023 - Present Prophecy Sensorlytics - Kolkata, India 06/2020 - 10/2021 <i>Project Manager — Condition-Based Maintenance</i> Freelance: 06/2020 - 10/2020, Full-time: 10/2020 - 11/2021 Research Scholar - IIT Kharagpur, India 07/2015 - 06/2020
PROJECTS	Data-driven machinery fault diagnosis: Applied machine learning to classify faults in bearings using publicly available dataset. [Link] Data-driven remaining useful life (RUL) prediction: Applied machine learning to predict remaining useful life of machinery using publicly available dataset. [Link] Open-source contribution: Contributed TensorFlow 2 code for Attention Mechanisms and Generative Adversarial Networks chapter of Dive into Deep Learning (D2L) book. [Link]
PUBLICATIONS	Sahoo, B. , & Mohanty, A.R., (2021). Multiclass Bearing Fault Classification Using Features Learned by a Deep Neural Network. In International Congress and Workshop on Industrial AI (pp. 405-414). Springer, Cham. [Link] [Code] Datta, B. N., & Sahoo, B. (2021). Machine Learning, Regression and Optimization. In Data Science and SDGs (pp. 177-197). Springer, Singapore. [Link] [Code]

Sahoo, B., & Mohanty, A. R. (2020). Feature Subset Selection Using Sparse Principal Component Analysis and Multiclass Fault Classification Using Selected Features. In Advances in Asset Management and Condition Monitoring (pp. 147-158). Springer, Cham. [\[Link\]](#) [\[Code\]](#)

PATENTS

Generating HTML document to include most relevant content sections of corresponding web page for printing (US Patent) (Filed. Application No.- PCT/US2025/046144)

TEACHING
EXPERIENCE

Teaching assistant for NPTEL MOOC on Machinery Fault Diagnosis and Signal Processing from Jan-Apr 2018.

Teaching assistant at IIT Kharagpur for Machinery Fault Diagnosis and Signal Processing, Machine Design, Machine Design Practice, Engineering Mechanics, Mechanics of Solids Laboratory, Engineering Drawing.